QUESTIONS 4

1. For the function $f(x) = \sqrt{1+x}$, let $x_0 = 0, x_1 = 0.6$, and $x_2 = 0.9$. Construct Lagrange interpolation polynomial of degree at most two to approximate $f(0.45)$, and compare the error.

2. Let $f(x) = \sin(\pi x)$.

- $x_0 = 1, x_1 = 1.25$, and $x_2 = 1.6$ are given. Construct Newton's Divided-Difference polynomial of degree at most two.
- $x_0 = 1, x_1 = 1.25, x_2 = 1.6$ and $x_3 = 2$ are given. Construct Newton's Divided-Difference polynomial of degree at most three.